Wildlife Rescue and Rehabilitation Management System

Object-Oriented Analysis and Design

Review – 2

Submitted by

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1. Object-Oriented Analysis (OOA)

1.1 Problem Statement

Citizens report injured or distressed wildlife. A coordinator dispatches field agents and tracks each rescue case. Veterinarians and caretakers examine, treat and monitor admitted animals until they are released or moved to permanent care. The system also supports volunteer scheduling, enclosure capacity tracking, inventory, donations, operational reporting and role-based access for every user.

1.2 System Actors

Actor	Role in the system
Citizen	Reports an injured/distressed animal. Only the report is stored, not the citizen.
Coordinator	Assigns field agents, tracks cases; manages staff, volunteers, inventory and donations.
Field Agent	Travels to the site, captures the animal and transports it to the centre.
Veterinarian	Diagnoses, prepares the treatment plan and decides release readiness.
Caretaker	Daily feeding, monitoring and behaviour observation of admitted animals.
Volunteer	Performs tasks and shifts assigned by the coordinator.
Donor	Contributes money or supplies to the centre.
Forest Department	External government authority that issues release permits.

1.3 Use Cases by Package

Package (Actors)	Use cases
1. Incident Reporting & Rescue Dispatch	Report distressed animal; log rescue request; assign rescuer; track rescue team; capture & transport animal (Citizen, Coordinator, Field Agent)
2. Animal Intake & Medical Care	Register animal; initial health assessment; diagnose; create treatment plan; administer treatment; schedule surgery; update medical records; monitor recovery; manage quarantine (Veterinarian, Caretaker)
3. Habitat & Daily Care	Assign enclosure; schedule feeding; track diet; monitor behaviour (Caretaker, Volunteer)
4. Release & Post-Release Tracking	Release-readiness assessment; coordinate permits; schedule & log release; transfer to sanctuary; GPS tracking (Veterinarian, Coordinator, Forest Dept.)
5. Staff & Volunteer Administration	Register staff/volunteers; assign shifts; track training; manage attendance (Coordinator, Volunteer)
6. Inventory & Asset Management	Medical and food supply inventory; track equipment/vehicles (Coordinator, Caretaker)
7. Donation & Reporting	Manage donations; generate operational reports (Donor, Coordinator)

1.4 Identifying Classes

Four complementary approaches were applied.

- **Noun Phrase (NP):** nouns in the problem statement become candidate classes; synonyms, vague terms and attributes are rejected.
- **Common Class Pattern (CC):** candidates are checked against people, places, things, events and records, exposing classes hidden in verbs (diagnose → MedicalRecord, treat → TreatmentPlan, monitor → HealthLog, release → ReleaseRecord).
- **Use-Case Driven (UC):** every class must take part in some use case; this is the only approach that yields boundary and control classes.
- **CRC:** each class is described by its responsibilities and the collaborators it needs to fulfil them.

Rejected candidates: Citizen (external actor; no data stored), centre (the system itself), occupancy (attribute of Enclosure), role (attribute of User), currentStatus (attribute of Animal) and "reports" (becomes the generateReport() operation).

Class	Type	NP	CC	UC	CRC	Key responsibility → collaborators
User	Entity (base)	✓	–	✓	–	Login, role-based access → all role classes
Coordinator	Role	✓	✓	✓	✓	Assign agents, manage volunteers/inventory/donations → RescueCase, Volunteer, Donation
FieldAgent	Role	✓	✓	✓	✓	Capture & transport, update case status → RescueCase, Animal
Veterinarian	Role	✓	✓	✓	✓	Diagnose, plan treatment, assess release → MedicalRecord, TreatmentPlan, ReleaseRecord
Caretaker	Role	✓	✓	–	✓	Feeding, monitoring, quarantine care → HealthLog, Enclosure
Volunteer	Role	✓	✓	✓	✓	Perform assigned tasks → Coordinator, Caretaker
Donor	Role	✓	✓	✓	✓	Make donations, view history → Donation
Animal	Core entity	✓	✓	✓	✓	Identity, species, status; links to all care records → Enclosure, MedicalRecord
Enclosure	Place	✓	✓	✓	✓	Capacity, occupancy, quarantine status → Animal, Caretaker
IncidentReport	Event	✓	✓	✓	✓	What/where/when of a sighting → RescueCase
RescueCase	Event	–	✓	✓	✓	Links report, agent and rescued animal → IncidentReport, FieldAgent, Animal
MedicalRecord	Record	–	✓	✓	✓	Medical history and diagnoses → Animal, Veterinarian
TreatmentPlan	Record	✓	✓	✓	✓	Medication, schedule, completion → Animal, Caretaker
HealthLog	Record	✓	✓	✓	✓	Daily weight, temperature, behaviour → Animal, Caretaker
ReleaseRecord	Record	✓	✓	✓	✓	Readiness decision, permit, outcome → Animal, Veterinarian
Donation	Event	✓	✓	✓	✓	Amount, date, purpose, status → Donor, Coordinator
Inventory / SupplyItem / Equipment	Things	–	✓	–	–	Stock levels, expiry, equipment maintenance → Coordinator
IncidentReportForm	Boundary	–	–	✓	–	Citizen-facing report entry form
IncidentController, RescueController, MedicalController	Control	–	–	✓	–	Coordinate the reporting, dispatch and medical use-case flows

1.5 Class Diagram

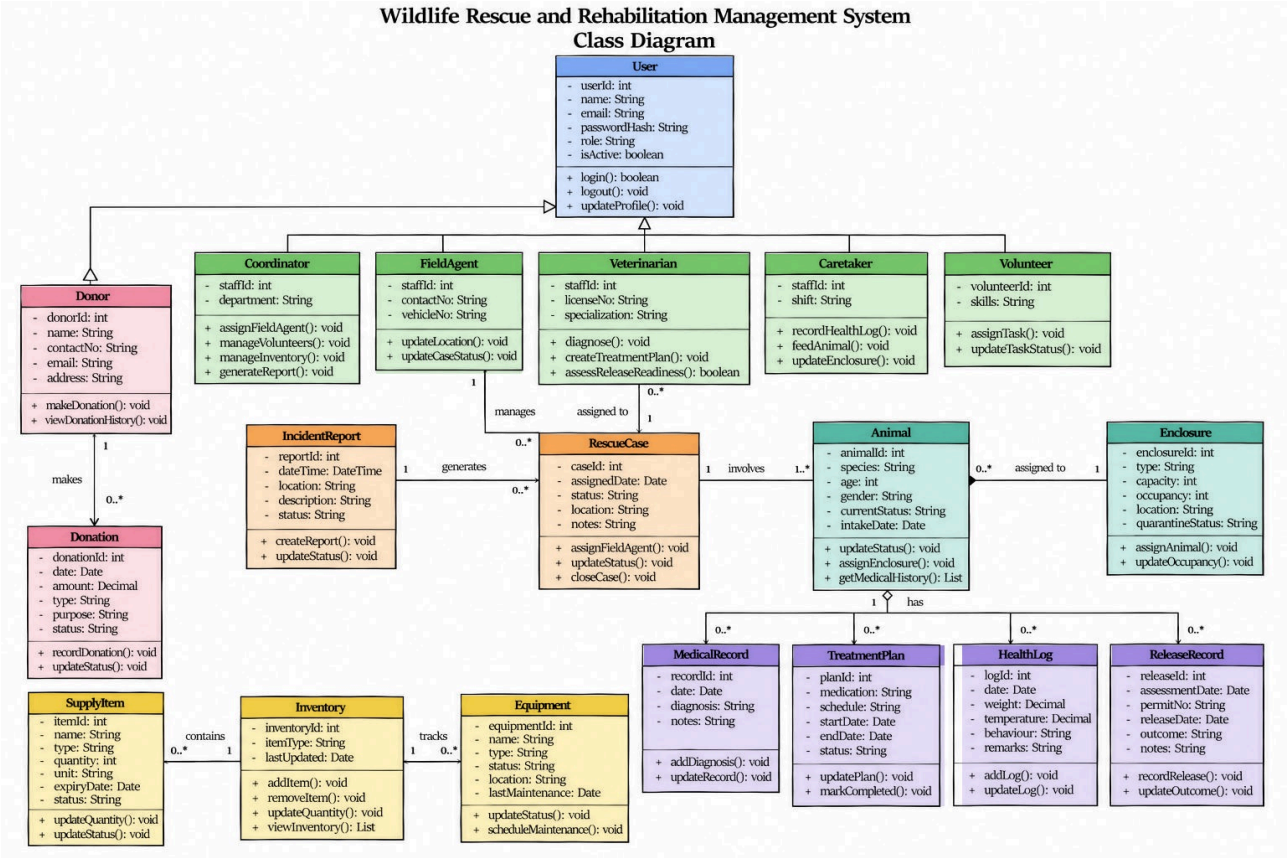


Figure: Class diagram of the Wildlife Rescue and Rehabilitation Management System

Colour key

- User (base class)
- Staff roles
- Donor and donation
- Incident and rescue case
- Animal and enclosure
- Medical and care records
- Inventory and equipment

2. Use Case Diagram

The use case diagram shows the main functions of the system and the actors who use them. Rescue and care staff are on the left, and the coordinator, volunteers, donors and the Forest Department are on the right.

- «include» – Create Treatment Plan always updates the medical record; Release Animal always needs a release permit.
- «extend» – Transfer to Sanctuary happens only when the release-readiness assessment finds the animal unfit for the wild.

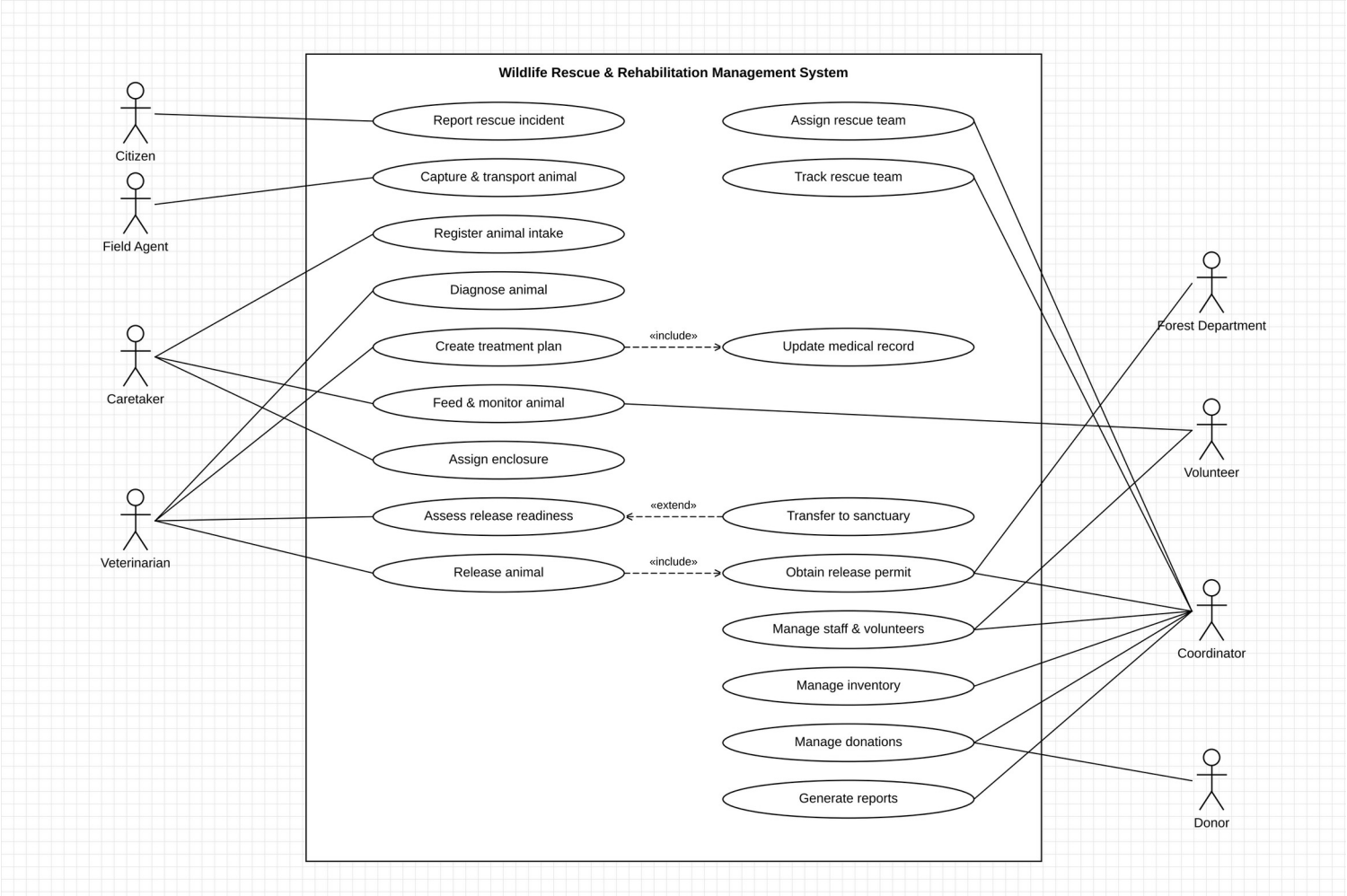


Figure 2.1: Use case diagram

2.1 Use Case Specifications

Field	Description
Use case	UC-01 Report Rescue Incident
Actors	Citizen (primary); Coordinator (notified)
Precondition	Citizen has access to the public reporting page or app.
Main flow	1. Citizen enters species, location, photos, description and contact details. 2. System validates the input. 3. System creates an incident record with status "New" . 4. System notifies the coordinator and shows the incident ID to the citizen.
Alternate flow	2a. Input invalid → system shows validation errors; citizen corrects and resubmits.
Postcondition	An incident with status "New" exists and the coordinator has been notified.

Field	Description
Use case	UC-02 Assign Rescue Team
Actors	Coordinator (primary); Field Agent
Precondition	Coordinator is logged in; incident is in status "New".
Main flow	1. Coordinator opens the incident. 2. System recommends available teams sorted by distance, species skill and availability. 3. Coordinator selects a team; system sends an assignment notification. 4. Field agent accepts; case status becomes "Assigned" and the reporter receives the ETA.
Alternate flow	4a. Team rejects → system reassigns to the next available team and informs the reporter. 4b. No team available → case is cancelled with a recorded reason.
Postcondition	The rescue case has an accepted field team and live tracking starts.

Field	Description
Use case	UC-03 Assess Release Readiness
Actors	Veterinarian (primary); Coordinator, Forest Department
Precondition	Animal is in status "Recovering" and flagged eligible for release.
Main flow	1. Veterinarian performs health check and behaviour test. 2. System stores the result and sets status "Release Pending". 3. If passed, a release plan is created and a permit is requested from the Forest Department.
Alternate flow	3a. Failed → animal returns to rehabilitation. 3b. Cannot survive in the wild → Transfer to Sanctuary (extend).
Postcondition	Animal is marked Ready for Release, returned to care, or transferred.

3. State Diagrams

State diagrams are drawn for the two classes whose behaviour changes most over time: Animal and RescueCase (created from an IncidentReport). Each state shows its entry/do activities; transitions are labelled event [guard] / action.

3.1 Animal

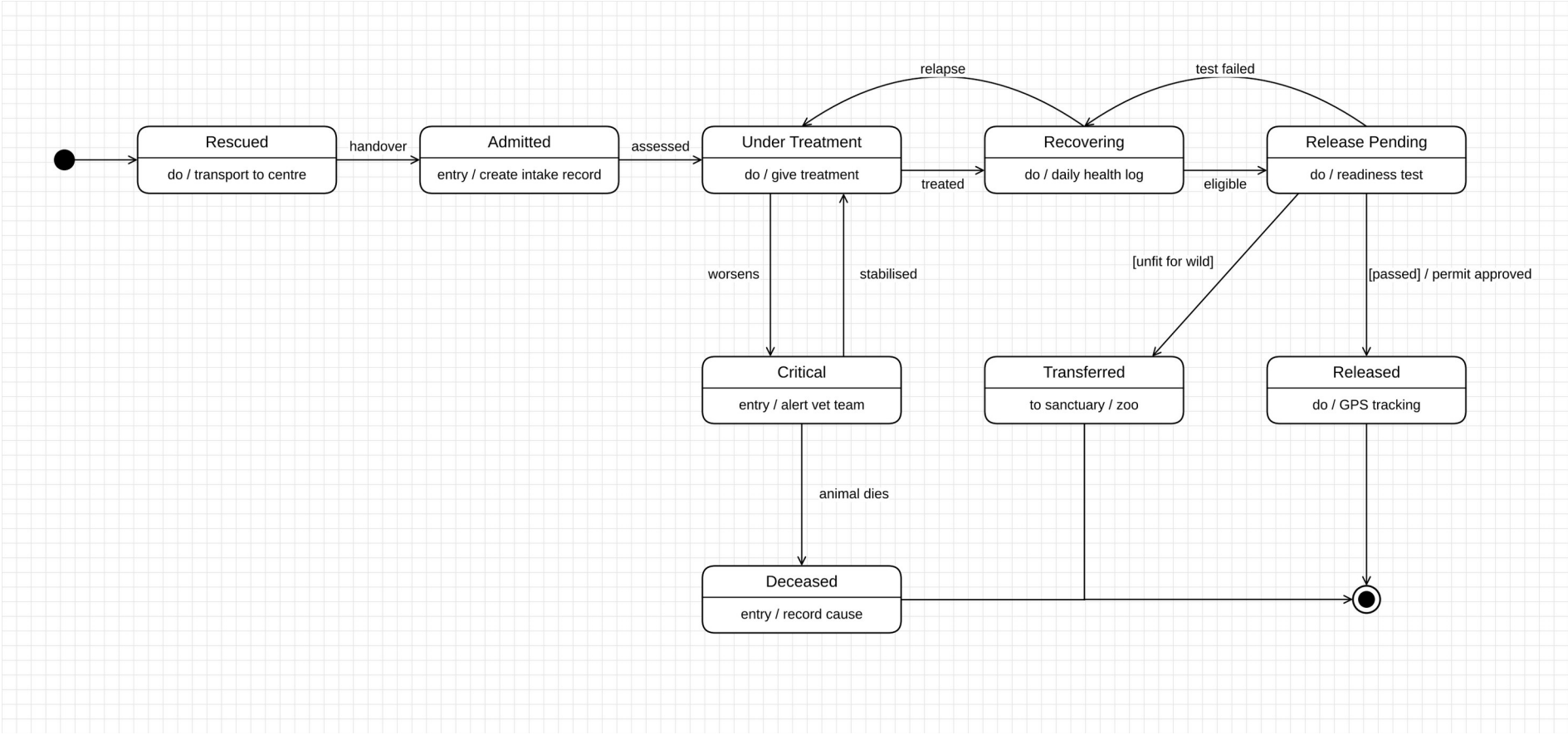


Figure 3.1: State diagram of Animal

The animal is rescued, admitted and put under treatment. It can become critical and return once stabilised, or recover and move to release pending. If the readiness test passes and the permit is approved it is released; if it cannot survive in the wild it is transferred to a sanctuary. The record also ends if the animal dies.

3.2 Incident Report / Rescue Case

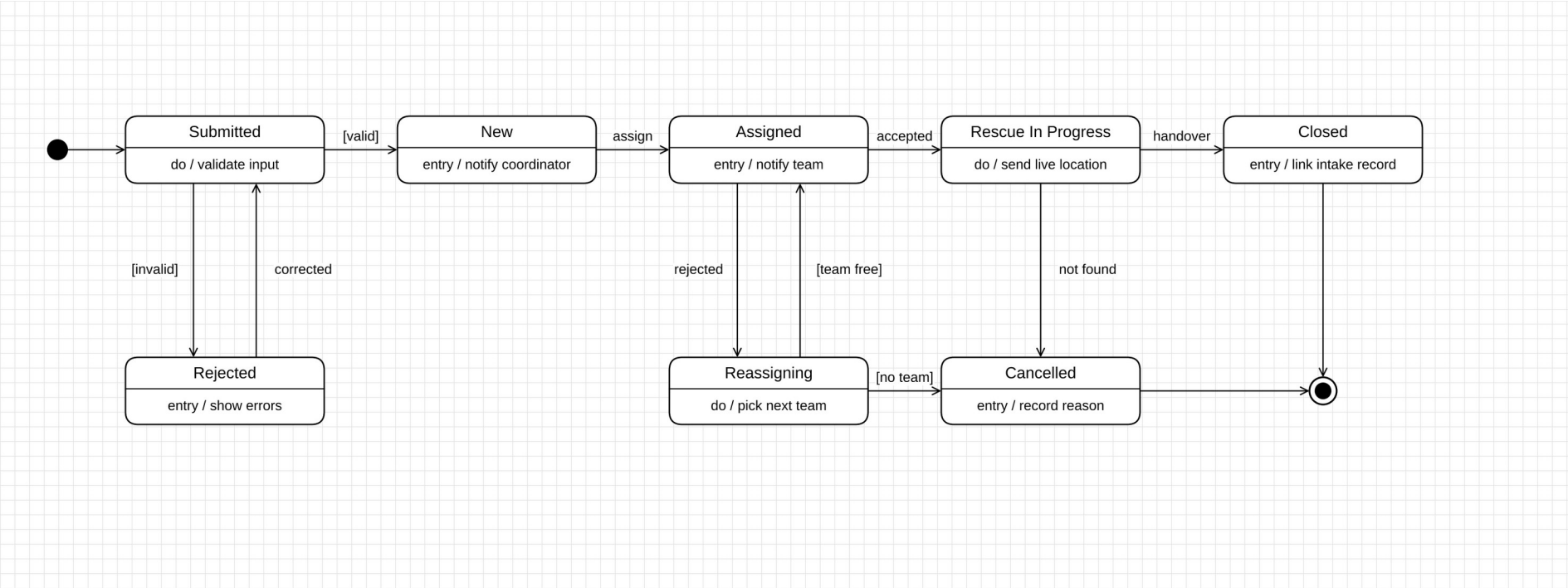


Figure 3.2: State diagram of RescueCase

A submitted report is validated; invalid reports are rejected until corrected. A valid report becomes a New case and is assigned to a team. If the team rejects, the case is reassigned. Once accepted the rescue is in progress and the case closes when the animal is handed over, or is cancelled if no team is free or the animal is not found.

4. Component Diagram

The system is divided into the following components. Client components run on user devices; service components run on the application and background-jobs servers (see the deployment diagram); external components are third-party systems reached over HTTPS.

4.1 List of All Components

Component	Type	Provided interface	Responsibility
Public Reporting UI	Client	–	Citizen submits incident reports
Rescue Field App	Client	–	Field agent GPS, photos, status updates
Staff Web Portal	Client	–	Veterinarian / caretaker work screens
Admin & Analytics Console	Client	–	Coordinator dashboards and administration
Volunteer & Donor Portal	Client	–	Volunteer shifts, donor contributions
Web Application / UI Gateway	Gateway	IWebAPI	Serves UI, routes API calls to services
Authentication Service	Service	IAuth	Login, sessions, role-based access
Incident Service	Service	IIncident	Validate and store incident reports
Rescue Dispatch Service	Service	IDispatch	Recommend, assign and reassign rescue teams
Location Tracking Service	Service	ILocation	Live rescue-team and post-release GPS
Animal Intake Service	Service	IIntake	Register admitted animals, intake records
Medical Service	Service	IMedical	Assessments, diagnoses, treatment plans
Rehabilitation Service	Service	IRehab	Treatment entries, progress metrics, status
Enclosure Service	Service	IEnclosure	Enclosure capacity, assignment, quarantine
Release Service	Service	IRelease	Release assessment, plans, permits
Staff & Volunteer Service	Service	IStaff	Registration, shifts, training, attendance
Inventory Service	Service	IInventory	Supplies, expiry, equipment maintenance
Donation Service	Service	IDonation	Record donations and payment status
Reporting Service	Service	IReport	Operational reports and analytics
Notification Service	Service	INotify	SMS / e-mail / in-app notifications
Database	Data	IPersistence	PostgreSQL WildCare schema
Background Jobs Scheduler	Service	IJobs	Scheduled tasks, reminders, report jobs
Maps & Geocoding API	External	IGeocoding	Maps, reverse geocoding
SMS / Email Gateway	External	ISMS	Message delivery
Payment Gateway	External	IPayment	Online donation payments
Forest Department Portal	External	IPermit	Release permit approval

4.2 Component Diagram – Incident Reporting & Rescue Dispatch Module

The Incident Reporting & Rescue Dispatch module is shown in detail. Incident Service and Rescue Dispatch Service provide the IIncident and IDispatch interfaces used by the client apps, and use the authentication, notification, database and maps interfaces of other components.

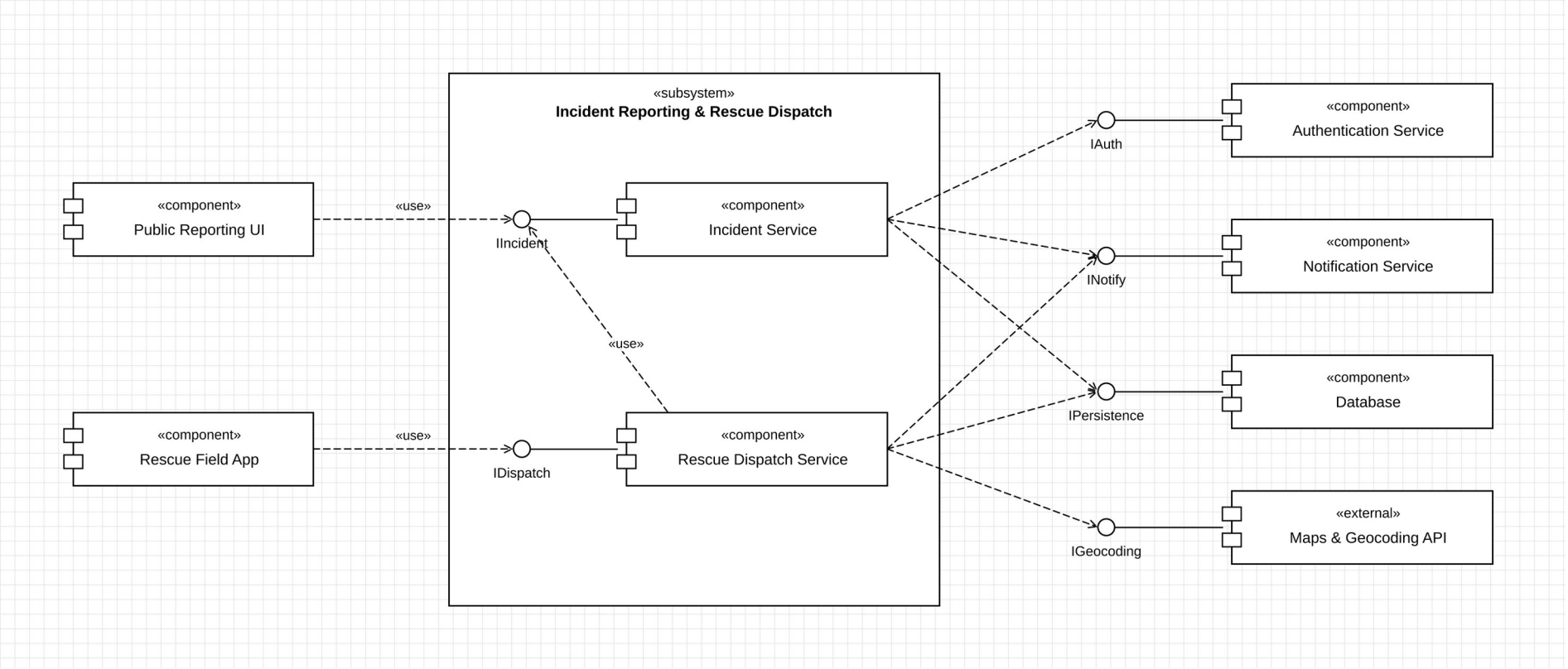


Figure 4.1: Component diagram – Incident Reporting & Rescue Dispatch module

5. Sequence Diagrams

5.1 Wildlife Rescue & Intake Management

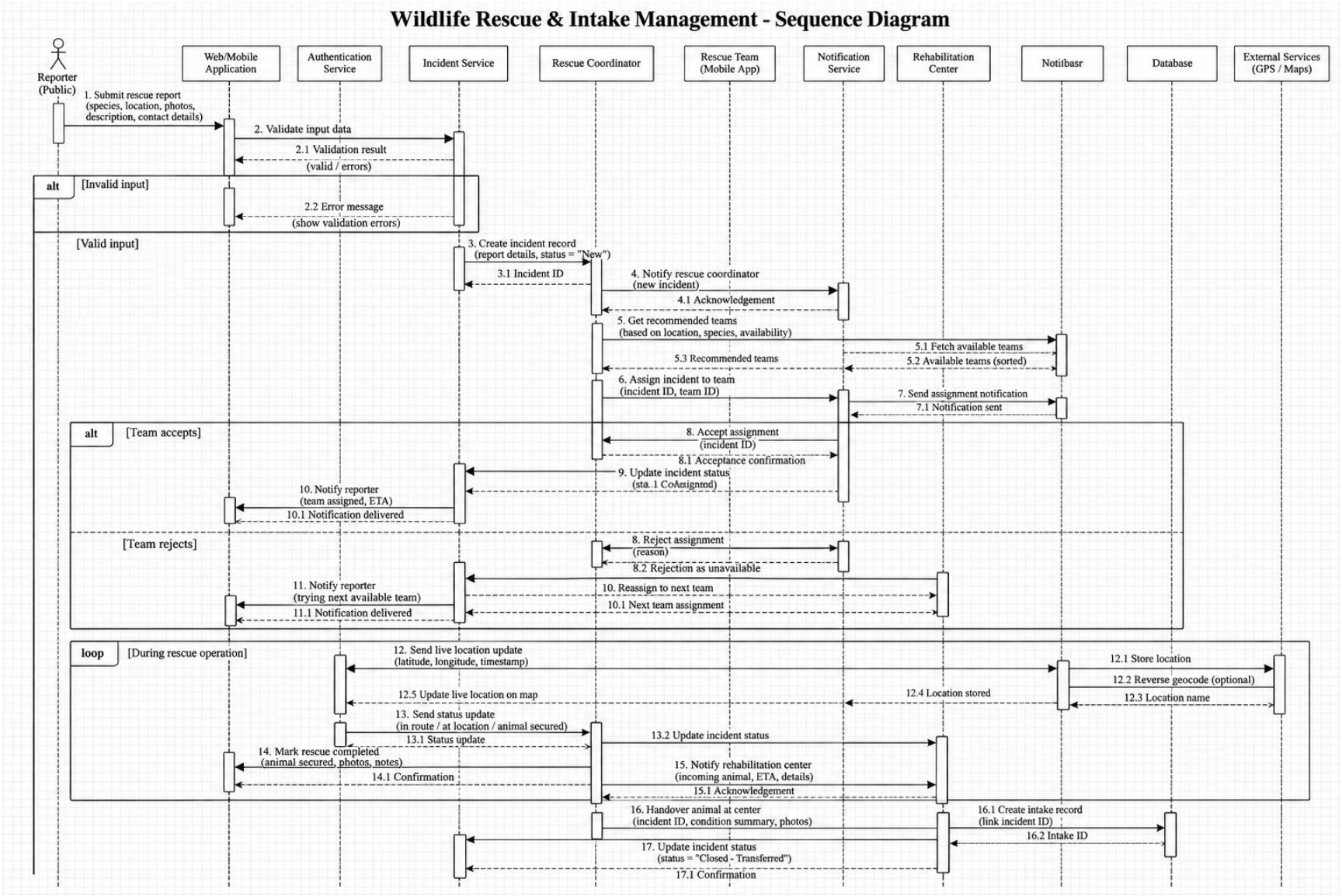


Figure 5.1: Report → dispatch → rescue → handover at the rehabilitation centre

5.2 Rehabilitation, Treatment & Release Management

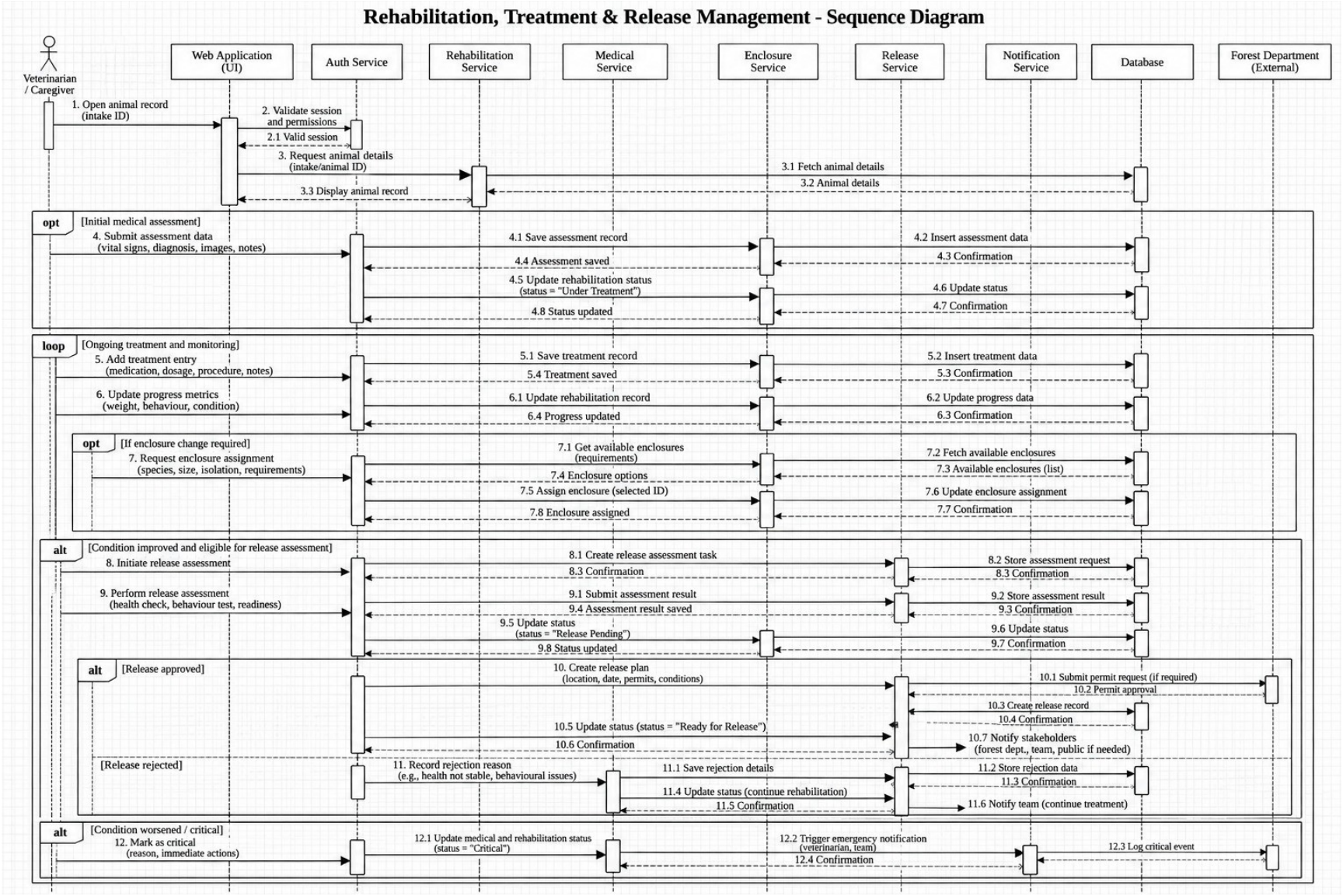


Figure 5.2: Assessment → treatment loop → release decision or critical escalation

6. Deployment Diagram

Each user device runs its client component and connects to the application server over HTTPS. All system components are deployed on the Wildlife Management Application Server, which stores data in the PostgreSQL database and talks to the external maps, SMS/email, payment and Forest Department services over HTTPS REST.

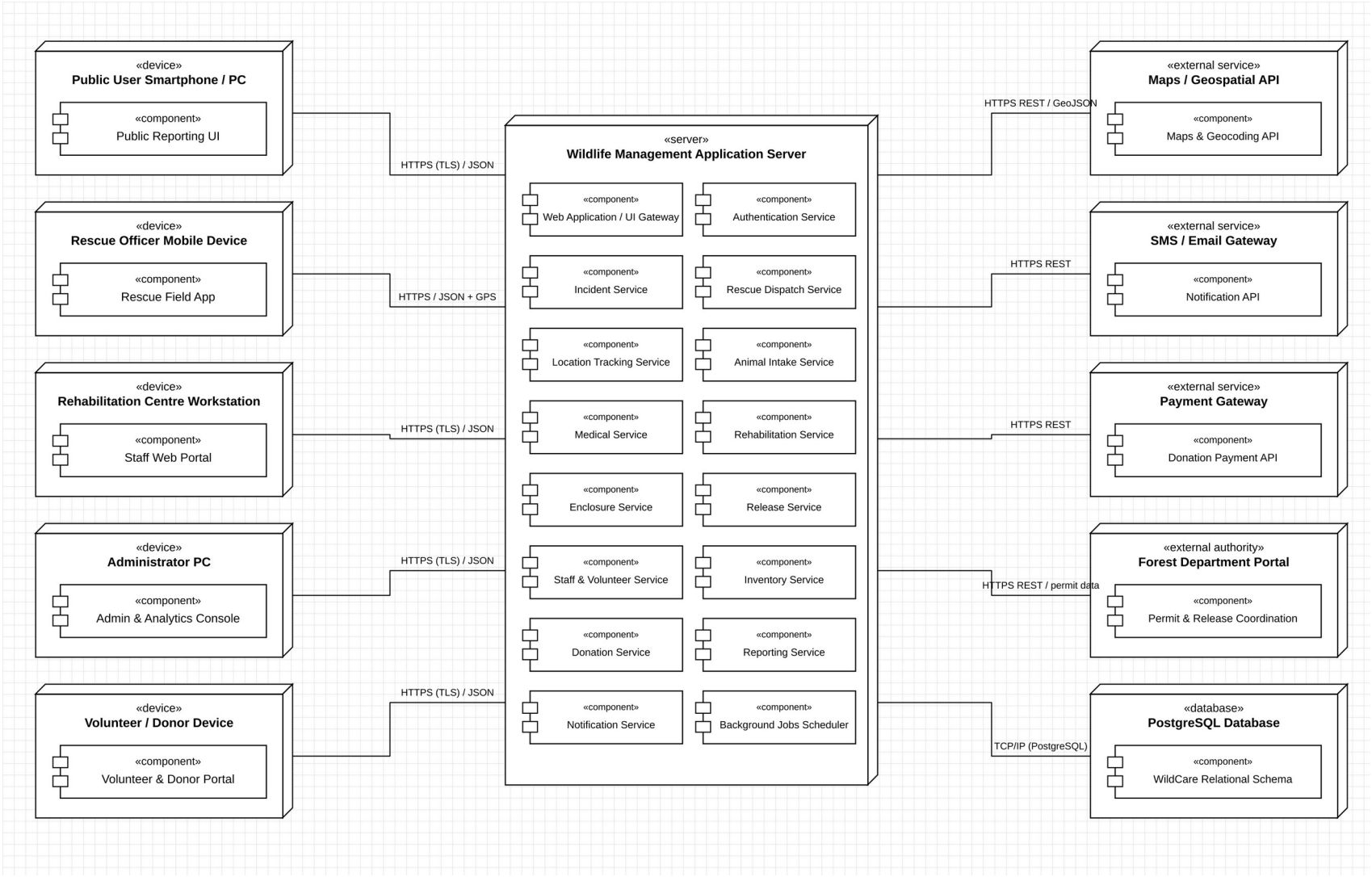


Figure: Deployment diagram

6.1 Deployment Nodes and Components

Node	Deployed components	Communication
Public User Smartphone / PC	Public Reporting UI	HTTPS (TLS) / JSON
Rescue Officer Mobile Device	Rescue Field App	HTTPS / JSON + GPS
Rehabilitation Centre Workstation	Staff Web Portal	HTTPS (TLS) / JSON
Administrator PC	Admin & Analytics Console	HTTPS (TLS) / JSON
Volunteer / Donor Device	Volunteer & Donor Portal	HTTPS (TLS) / JSON
Wildlife Management Application Server	Web Application / UI Gateway, Authentication, Incident, Rescue Dispatch, Location Tracking, Animal Intake, Medical, Rehabilitation, Enclosure, Release, Staff & Volunteer, Inventory, Donation, Reporting and Notification Services, Background Jobs Scheduler	–
PostgreSQL Database	WildCare Relational Schema	TCP/IP (PostgreSQL)
Maps / Geospatial API	Maps & Geocoding API	HTTPS REST / GeoJSON
SMS / Email Gateway	Notification API	HTTPS REST
Payment Gateway	Donation Payment API	HTTPS REST
Forest Department Portal	Permit & Release Coordination	HTTPS REST / permit data

7. Conclusion

The analysis identified eight actors, seven functional packages and twenty domain classes for the Wildlife Rescue and Rehabilitation Management System. The design models describe it from complementary views: the use case diagram fixes scope, state diagrams capture the life cycles of Animal and RescueCase, sequence diagrams show how components cooperate during rescue and rehabilitation, the component diagram organises the services behind clear interfaces, and the deployment diagram places every component on physical nodes. Together they form a consistent basis for implementation.

Team Contribution

Member	Register No.	Contribution
Pooja Reddy	25MIS0194	Use case diagram and specifications; state diagrams
Yogansh Palukuru	25MIS0262	Component diagram and component list; sequence diagrams
Vyom Gupta	25MIS0494	Deployment diagram; condensed OOA document

8. Implementation Screenshots

The design in this review was implemented as a working web application and deployed online. The application is built with Streamlit and a SQLite database, with role-based access for the actors identified in Section 1. The screenshots below show the public landing page and the citizen reporting page, which realise use case UC-01 (Report Rescue Incident).

8.1 Public Landing Page

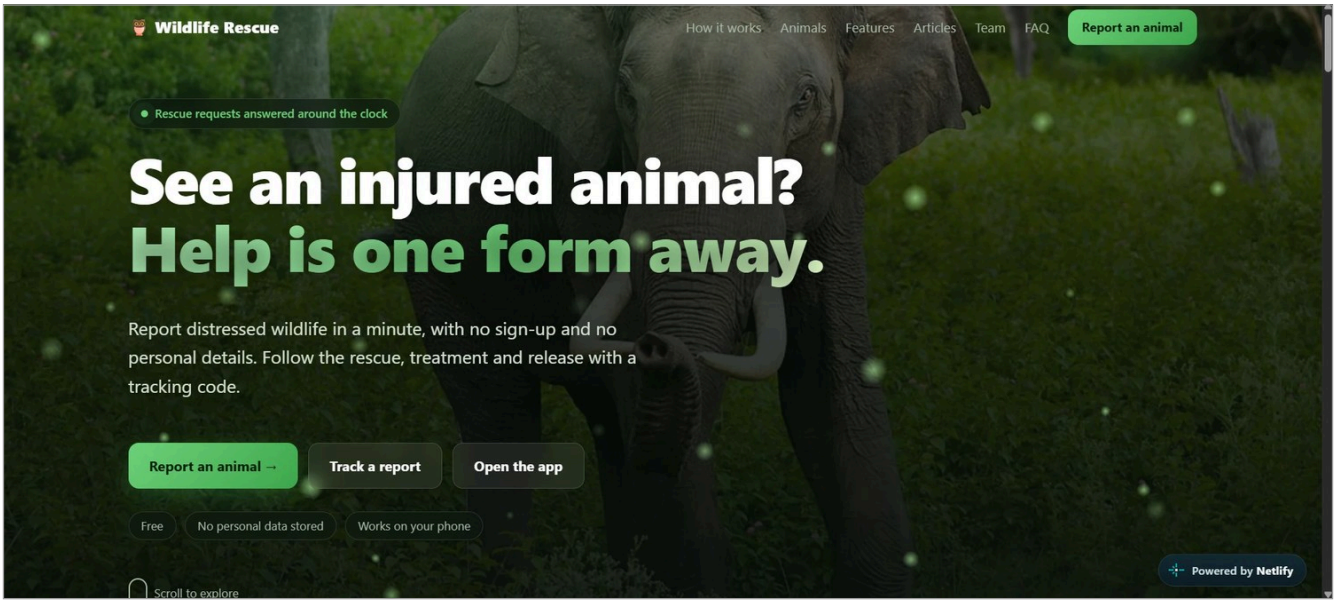


Figure 8.1: Landing page - hero section with report, track and open-app actions

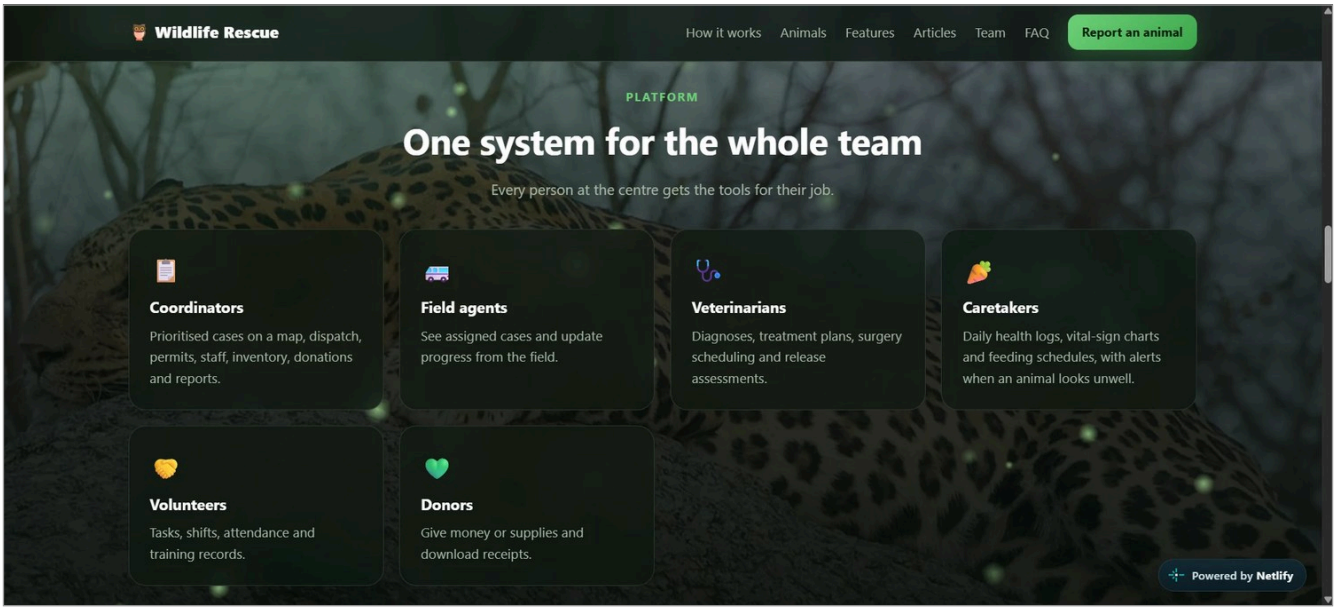


Figure 8.2: Landing page - one system for every actor (coordinator, field agent, veterinarian, caretaker, volunteer, donor)

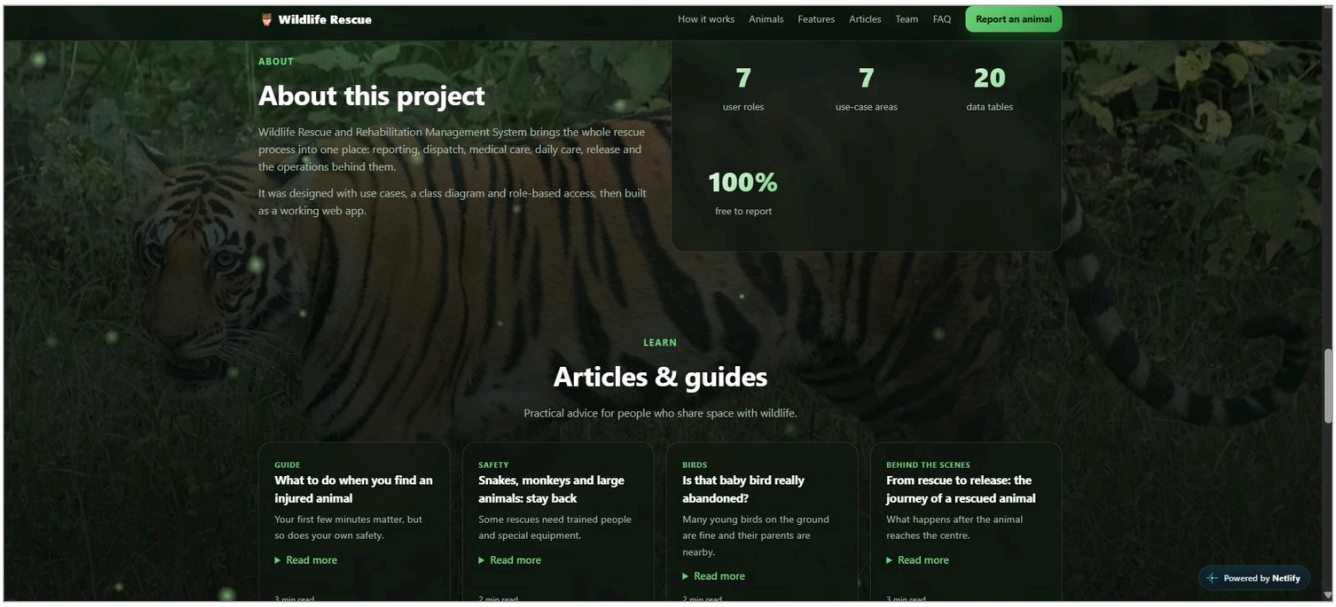


Figure 8.3: Landing page - about the project and educational articles

8.2 Citizen Reporting (UC-01)

A member of the public can report an injured or distressed animal without creating an account. No personal data is stored. The report records the location, species, urgency and an optional photo and map position, and returns a tracking code so the citizen can follow the rescue case.

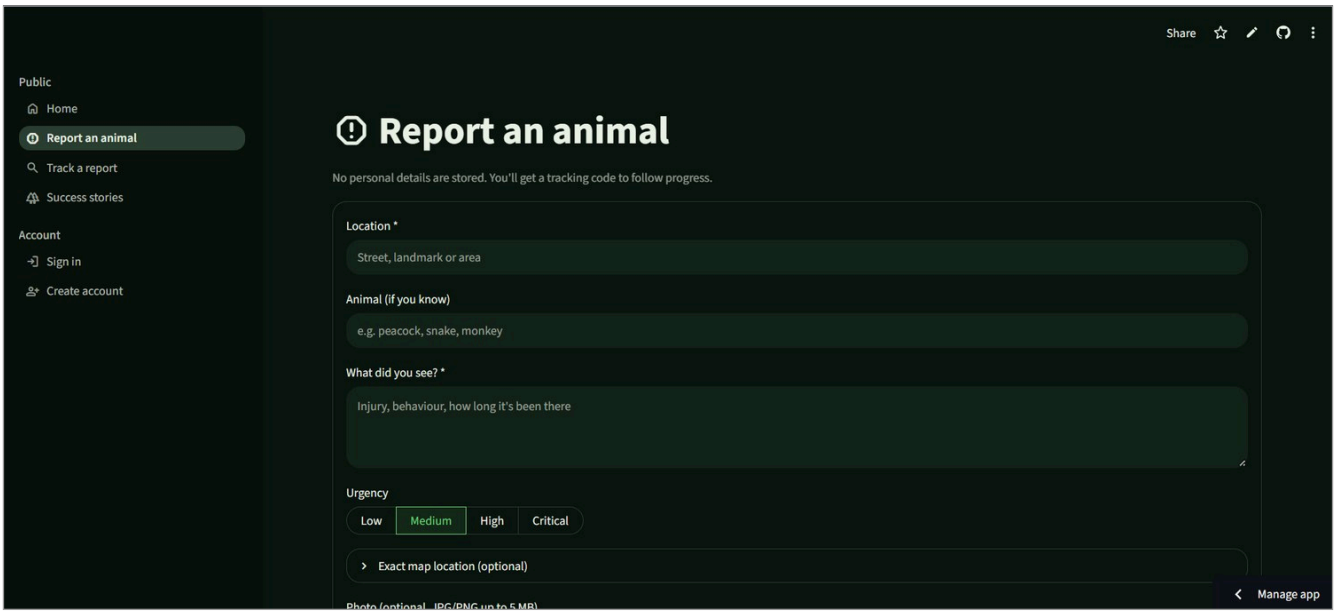


Figure 8.4: Report an animal form in the application (dark green theme)